Project Report: Atmos Full-Stack Weather Application

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About PM Accelerator

PM Accelerator is a product management training program built to help aspiring and early-career PMs develop real, job-ready skills. Through hands-on projects, mentorship from industry professionals, and a strong peer community, PMA bridges the gap between theoretical learning and practical product work. Their mission is to accelerate careers by giving people the tools, experience, and community they need to break into and grow within product management.

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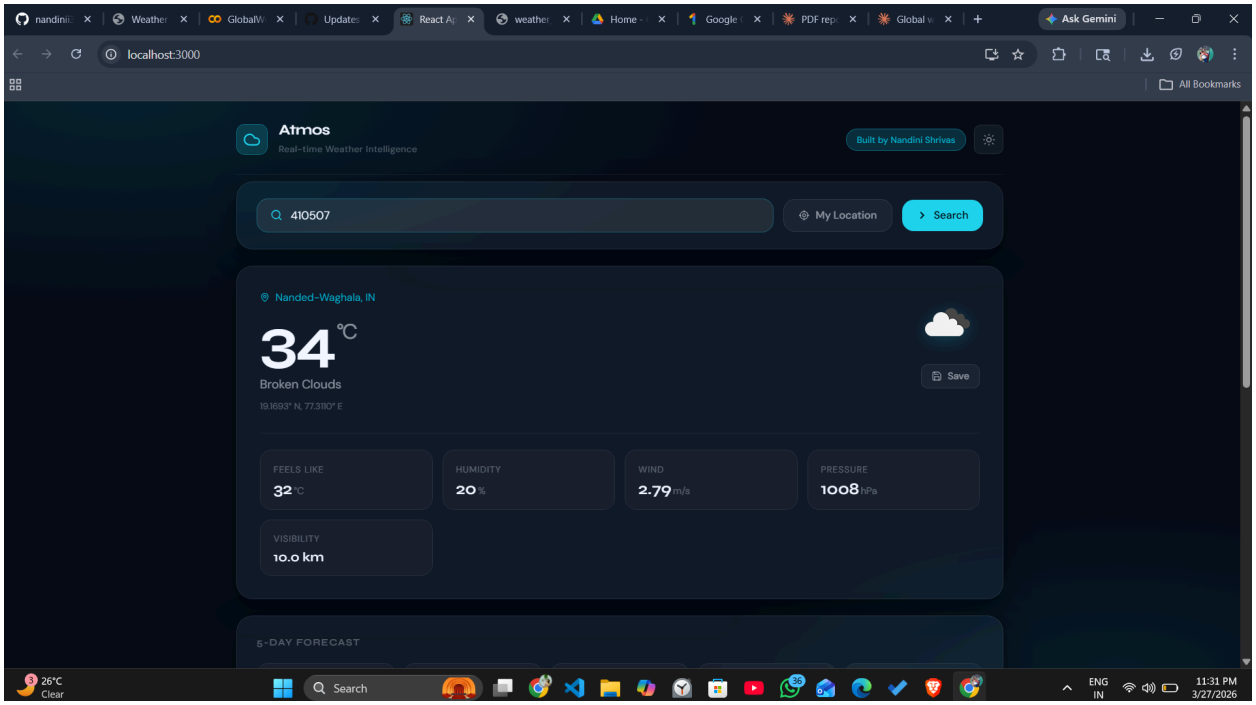
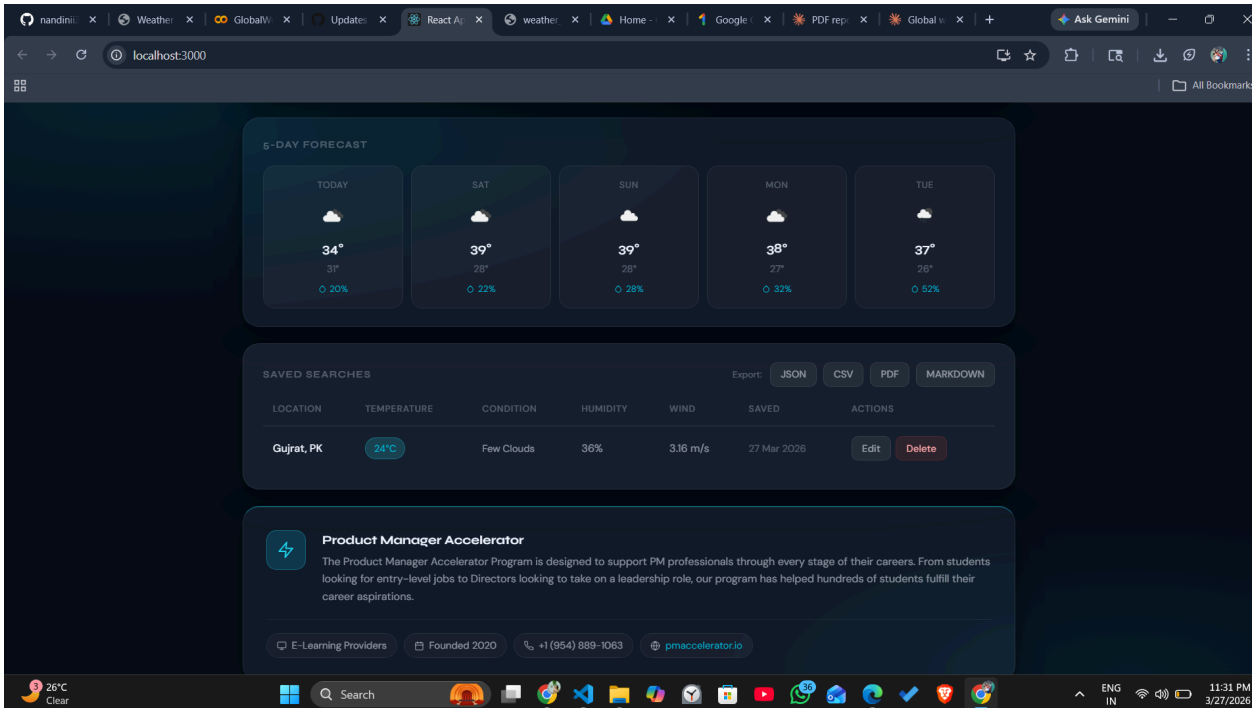
1. Executive Summary

- **Objective:** To build a responsive, feature-rich weather dashboard with integrated CRUD capabilities and third-party media integrations (Google Maps & YouTube).
- **Key Achievement:** Successfully implemented a "Glassmorphism" UI that maintains performance while handling multiple asynchronous API streams.





2. System Architecture

- **Frontend (The Face):** React.js handles the state management (using Hooks like useState and useEffect) and provides a seamless, single-page experience.
- **Backend (The Brain):** Node.js and Express manage the RESTful API routing, environment variables for security, and data transformation for exports.
- **Database (The Memory):** MongoDB Atlas stores user history, allowing for persistent data across sessions.

3. Feature Deep-Dive

A. Real-Time Data Integration

- **OpenWeatherMap:** Fetches current conditions and 5-day forecasts.
- **Geolocation API:** Uses browser-level permissions to provide localized weather instantly.

B. CRUD Operations & Persistence

- **Create:** Users can save specific weather snapshots with custom notes.
- **Read:** A historical table fetches stored data from MongoDB.
- **Update/Delete:** Full management of the history log with validation (e.g., preventing "End Date" from being earlier than "Start Date").

C. Enhanced User Experience (UX)

- **Media Integration:** Using the YouTube Data API and Google Maps Embed API to provide a "travel-ready" context for searched locations.
- **Data Portability:** Implementation of multi-format exports (JSON, CSV, PDF, Markdown) to show backend versatility.

4. Technical Challenges & Solutions

Challenge	Solution
API Latency	Implemented loading states and graceful error handling to prevent UI freezing during slow network calls.
Responsive Design	Used CSS Media Queries and Flexbox/Grid to ensure the Glassmorphism effect remained legible on mobile (420px).
Security	Leveraged .env files to mask sensitive API keys and database credentials from the frontend.

5. API Documentation (Summary)

- **GET /api/weather/live/:location**: The core "Search" engine.
- **POST /api/weather**: The bridge between the API and the Database.
- **GET /api/export/:format**: The utility service for data management.

6. Design Philosophy (The "PM" Perspective)

- **Glassmorphism:** Chosen for a modern, "airy" feel that mimics the atmosphere (Atmos).
- **Dark/Light Mode:** Included for accessibility and user preference.
- **Input Validation:** Essential for maintaining data integrity in the database.